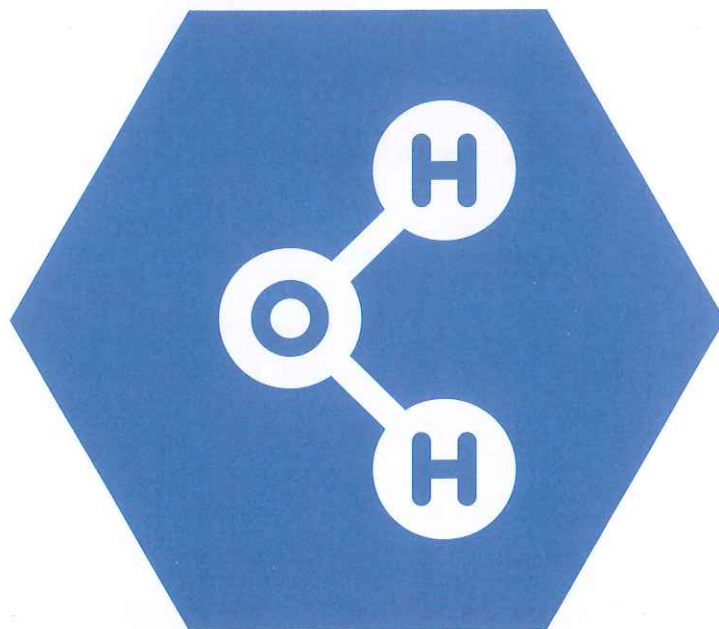




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THE UNIVERSITY OF NEW SOUTH WALES
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PAPER A



2013 ICAS

International Competitions
and Assessments for Schools

SCIENCE

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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

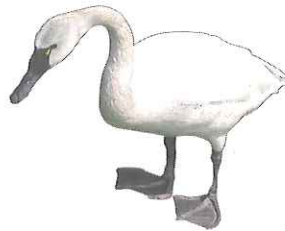
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

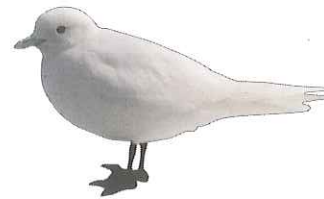
1. The pictures show some birds.



duck



swan



gull

Which part of these birds would best identify the gull from the others?

- (A) colour (B) eyes (C) feet (D) mouth

2. Eric sent four hand signals to his friend. He sent three of the signals with the same hand. He sent one signal using his other hand.

Which signal did he send with his other hand?

(A)



(B)



(C)

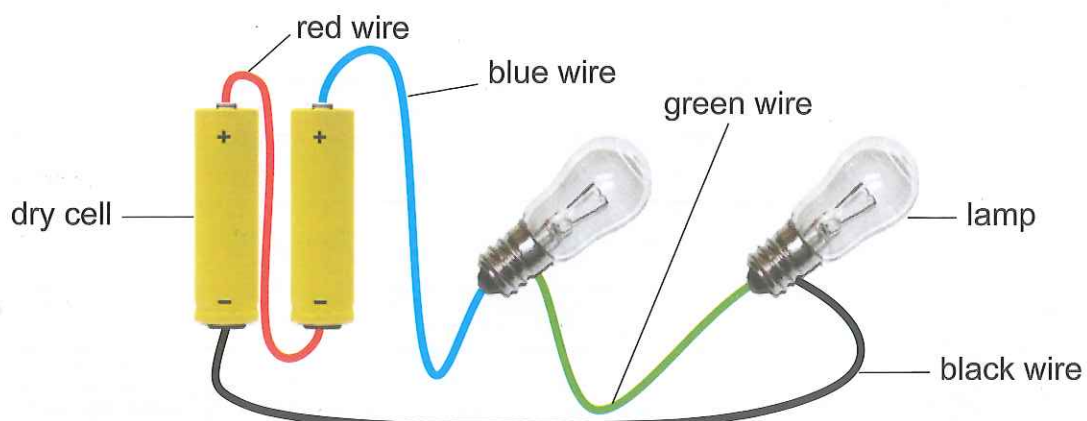


(D)



3. Mark made an electrical circuit using two dry cells and two lamps. He also used four wires, each of which had a different colour plastic coating.

The picture shows the circuit that he made.

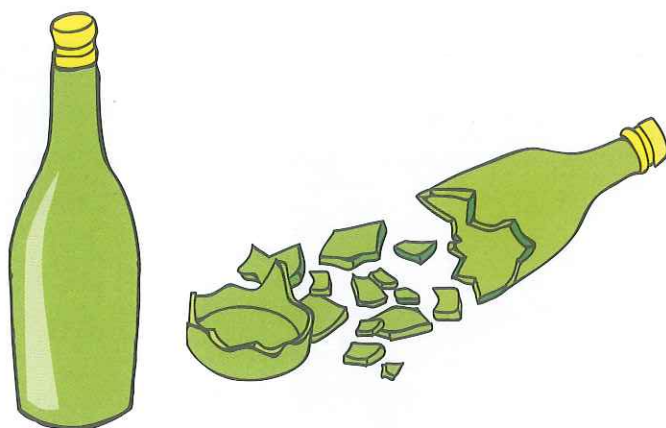


What is the colour of the wire that connects the two lamps to each other?

- (A) red (B) blue (C) black (D) green

4. Liam was carrying an empty glass bottle. He dropped the bottle and it broke.

The drawing shows the bottle before and after Liam dropped it.



What is different about the glass bottle after he dropped it?

- (A) the amount of glass
- (B) the shape of the bottle
- (C) the colour of the bottle
- (D) the material in the glass

5. Stan collected some rocks on his last holiday.

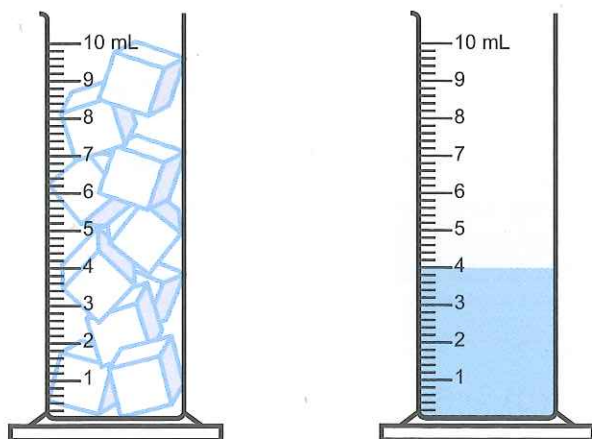
His favourite rock is the one that is smooth, rounded and has a light stripe.

Which of these rocks is Stan's favourite?



6. Sam put some ice cubes in a measuring cylinder. She left the measuring cylinder until all the ice cubes had melted.

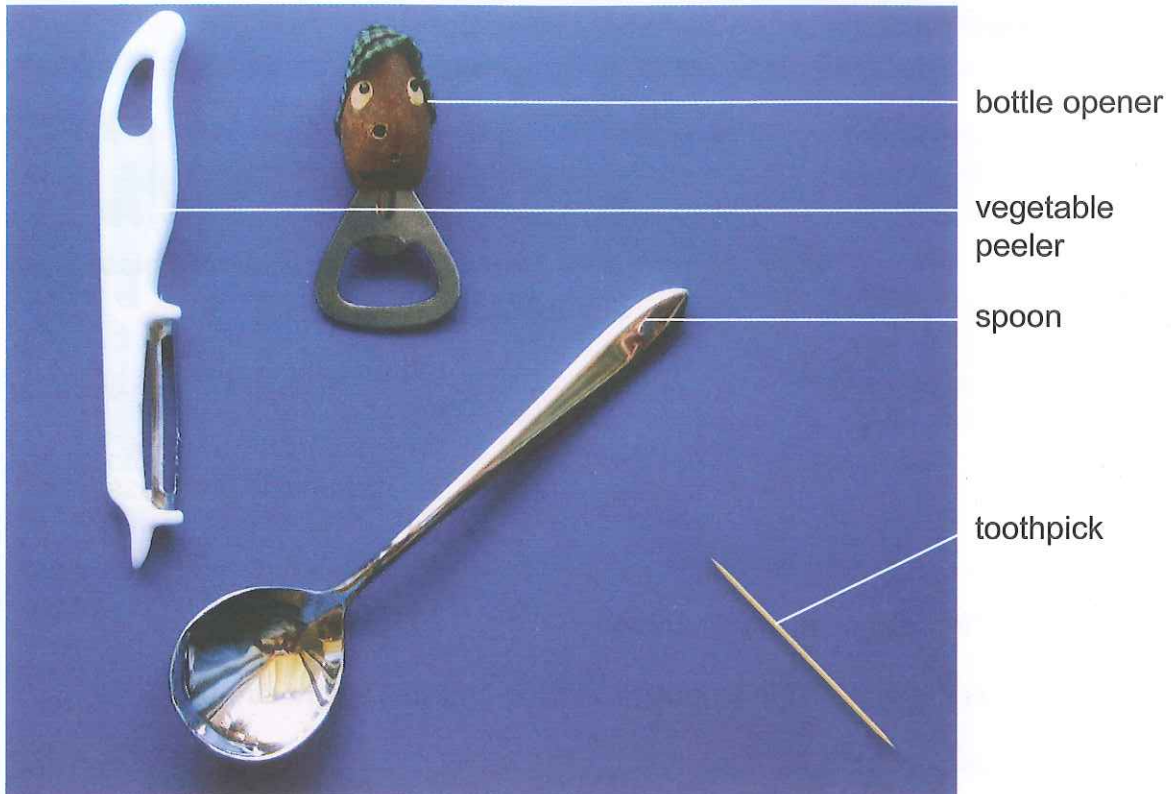
The diagrams show the measuring cylinder before and after the ice had melted.



How much water came from the ice cubes?

- (A) 10 mL
- (B) 6 mL
- (C) 4 mL
- (D) 1 mL

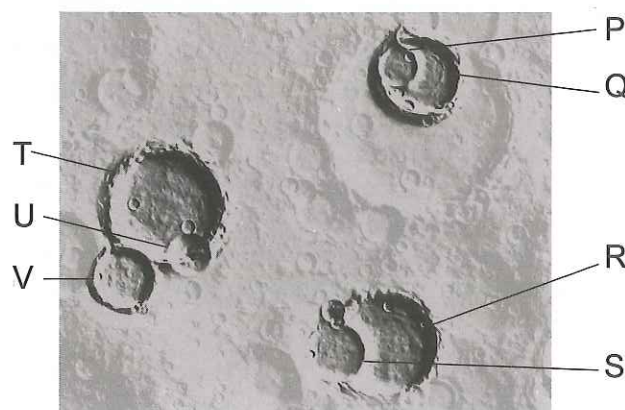
7. Jack took a photograph of some objects he found in the kitchen.



Which option lists the objects in order of length from shortest to longest?

- (A) toothpick, bottle opener, vegetable peeler, spoon
- (B) toothpick, bottle opener, spoon, vegetable peeler
- (C) bottle opener, vegetable peeler, toothpick, spoon
- (D) bottle opener, toothpick, vegetable peeler, spoon

8. The diagram represents some craters on the surface of the Moon.



Which of these craters is the widest?

- (A) Q
- (B) R
- (C) S
- (D) T

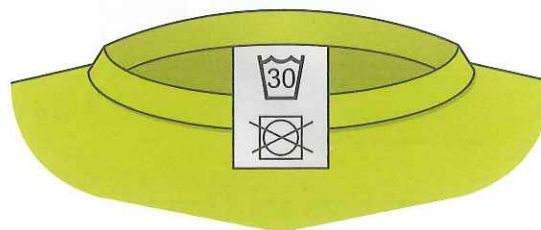
9. Most pieces of clothing have a label with symbols that show how to clean it.

This chart shows the meaning of some of these symbols.

Fabric Care Symbols

MACHINE WASH	TUMBLE DRY	DRY	IRON	DRY CLEAN
TEMPERATURE	HEAT SETTING	TEMPERATURE	TEMPERATURE	
30 Cool/Cold	No Heat	Line Dry/ Hang to Dry	Low	Dry Clean
40 Warm	Low	Drip Dry	Medium	Do Not Dry Clean
60 Hot	Medium	Dry Flat	High	
CYCLE	OTHER	OTHER	OTHER	
Normal	Any Heat	Do Not Wring	No Steam	
Permanent Press	Do Not Tumble Dry		Do Not Iron	
Delicate/Gentle				

The diagram shows the label attached to Sean's shirt.



Using these instructions, how should Sean clean his shirt?

- (A) wash in cool water; do not dry clean
- (B) wash in warm water; do not dry clean
- (C) wash in cool water; do not tumble dry
- (D) wash in warm water; do not tumble dry

10. John wants to reduce the amount of sugar and sodium in his diet and increase the amount of protein.

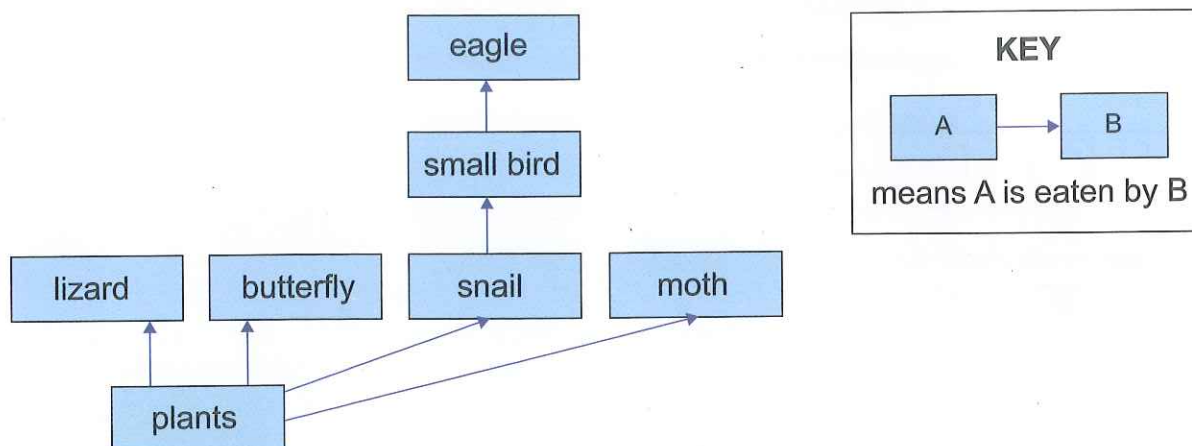
The table shows nutritional information for four foods in his diet.

	Chips	Lollies	Steak	Nuts
Protein (per 100 g)	7 g	0 g	26 g	27 g
Carbohydrate (per 100 g)	47 g	63 g	0 g	9 g
Sodium (per 100 g)	650 mg	110 mg	48 mg	500 mg
Sugar (per 100 g)	1 g	51 g	0 g	5 g

From the table, which food should John keep in his diet?

- (A) chips
- (B) lollies
- (C) steak
- (D) nuts

11. This is an example of a food web.

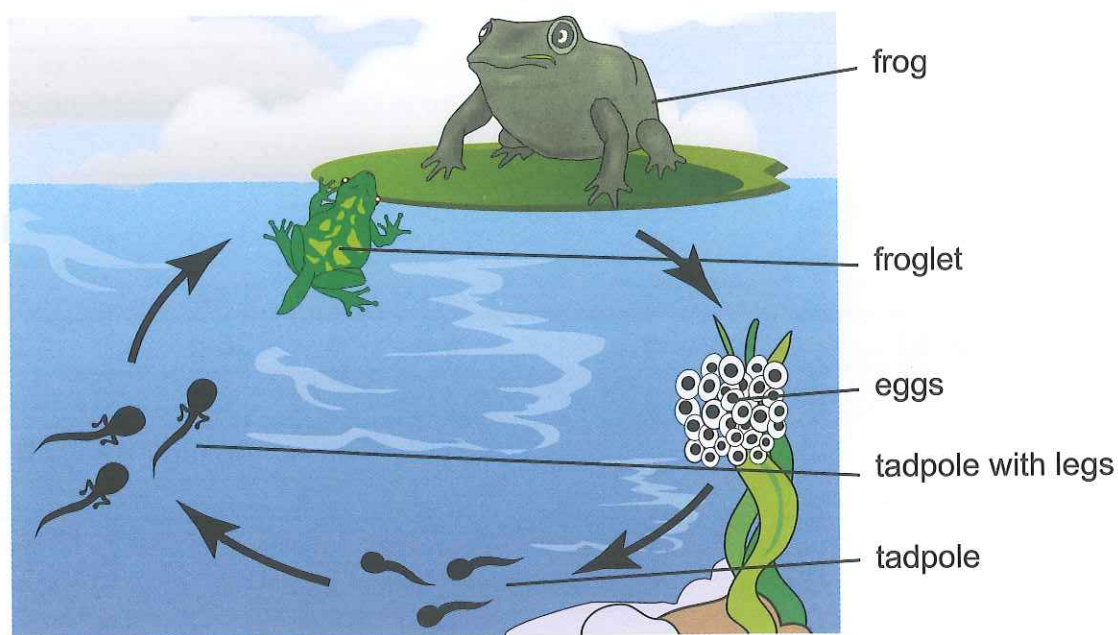


All of the small birds are killed off by a disease.

Which other group decreases in numbers as a result?

- (A) eagle (B) lizard (C) snail (D) moth

12. The diagram shows the stages in the life cycle of a frog. Each stage is drawn to its actual size.



At which stage of the frog's life cycle is it 1.5 cm in length?

- (A) tadpole (B) tadpole with legs
(C) froglet (D) frog

13. A student had four bricks. The bricks were the same size but they were made of different materials.



The student placed the bricks into four identical containers each with five litres of water and left them there. After 24 hours the student removed each brick from its container and measured the amount of water that remained in each container.

What question will the student be able to answer with the result?

- (A) Which brick is the strongest?
 - (B) Which brick weighs the most?
 - (C) Which brick absorbs the most water?
 - (D) Which brick cannot be used to build a house?
14. Some students measured how long it took carrot seeds to sprout at different temperatures.

The table shows their results.

Temperature (°C)	Days taken to sprout
22	17
24	15
26	13

Based on the table, how long will it take carrot seeds to sprout when the temperature is 28°C?

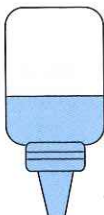
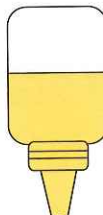
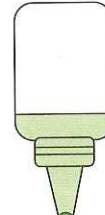
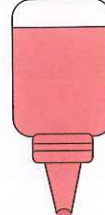
- (A) 9 days
- (B) 11 days
- (C) between 13 and 17 days
- (D) more than 17 days

15. Detergent works by breaking up oil into very small droplets.

Frank tested four brands of dishwashing detergent. He filled four identical dropper bottles with different brands of detergent. For each test, he added drops of one brand of detergent to some oil, one drop at a time. He did this until all the oil had broken up into droplets.

Frank used a teaspoon of oil for each test.

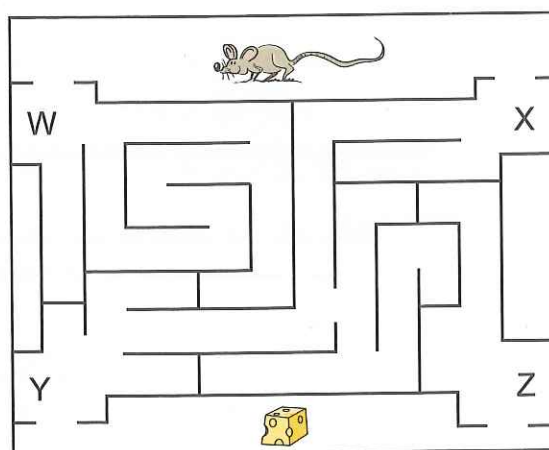
The table shows the dropper bottles at the end of the test for the four brands of detergent.

Test	1	2	3	4
Brand	Perfecto	Clean Up	Dish Lo	Sudsy
Dropper bottle at end of test				

Which brand of dishwashing detergent was the best at breaking up the oil?

- (A) Perfecto (B) Clean Up (C) Dish Lo (D) Sudsy

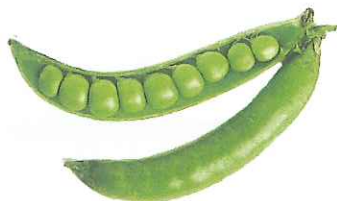
16. Mickey the mouse wants to make his way from one side of this maze to the other side so that he can reach the cheese.



Where must Mickey enter the maze and where must he leave the maze to reach the cheese?

	Enter at	Leave at
(A)	W	Y
(B)	W	Z
(C)	X	Y
(D)	X	Z

17. These photographs show different seed pods. Each pod has a different number of seeds inside it.



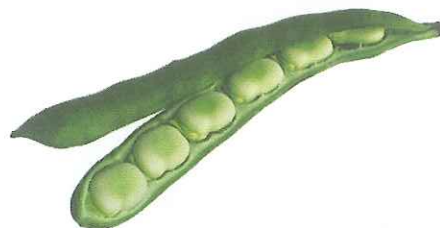
peas



bitter beans

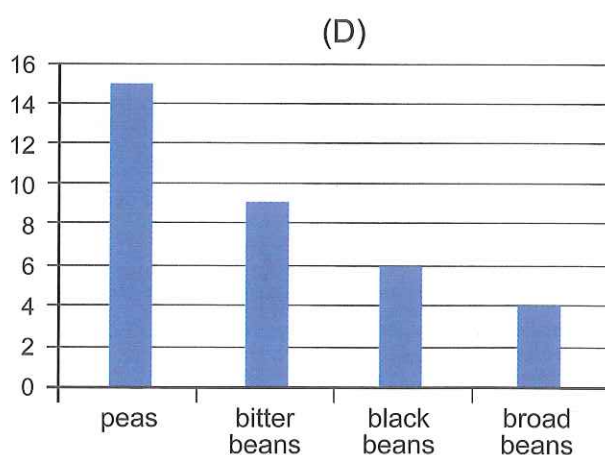
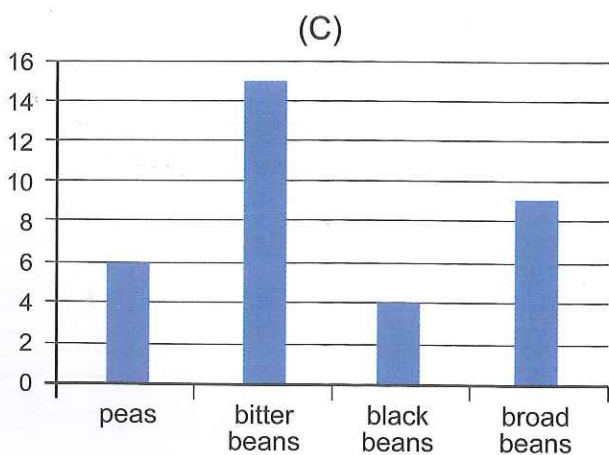
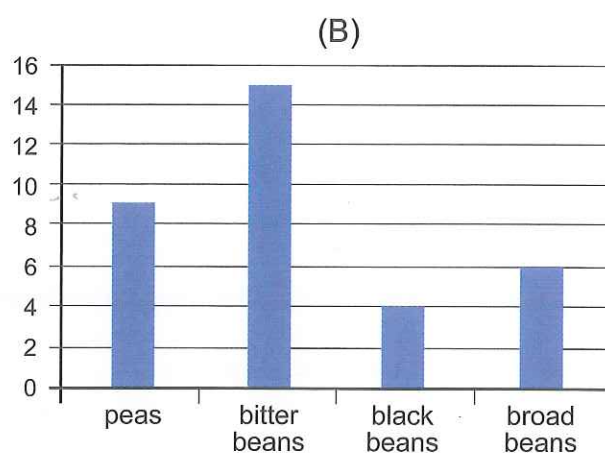
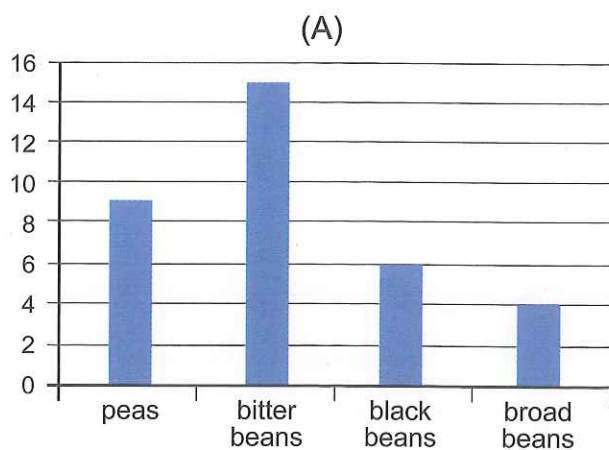


black beans



broad beans

Which graph shows the number of seeds in each pod?



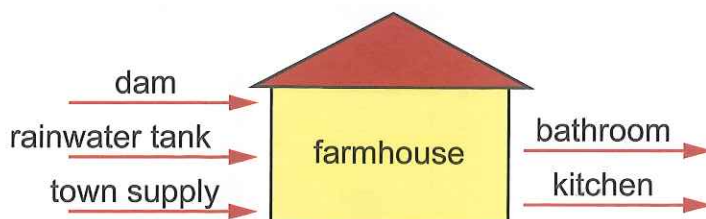
18. These photographs show bananas at different stages of ripening. As they ripen, the bananas change colour from green to yellow and then black.



Which is the correct sequence of bananas from least ripe to most ripe?

- (A) 2, 1, 4, 5, 3 (B) 2, 4, 1, 3, 5
(C) 2, 1, 4, 3, 5 (D) 2, 4, 1, 5, 3

19. The diagram shows the water that enters and leaves Sam's farmhouse. The arrows show the direction of the water flow.



Sam tested the quality of the water entering and leaving her farmhouse.

The table shows Sam's test results.

Water from	Colour	Taste/smell	Water clarity
dam	brown	earthy	clear
rainwater tank	colourless	none	clear
town supply	colourless	earthy	clear
bathroom	grey	soapy	cloudy
kitchen	grey	soapy	cloudy

What colour is the water that flows out of the farmhouse?

- (A) grey (B) brown
(C) clear (D) cloudy

For questions 20 and 21 use the information below.

Cali wanted to find out what substances look like when they are mixed with water.

The table lists her results.

Substance mixed with water	What colour is the mixture?	Can you see through the mixture?
milk	white	no
mud	brown	no
salt	no colour	yes
sugar	no colour	yes
tea	brown	yes

20. After mixing with water, which two substances can Cali **NOT** tell apart using only the results of her investigation?

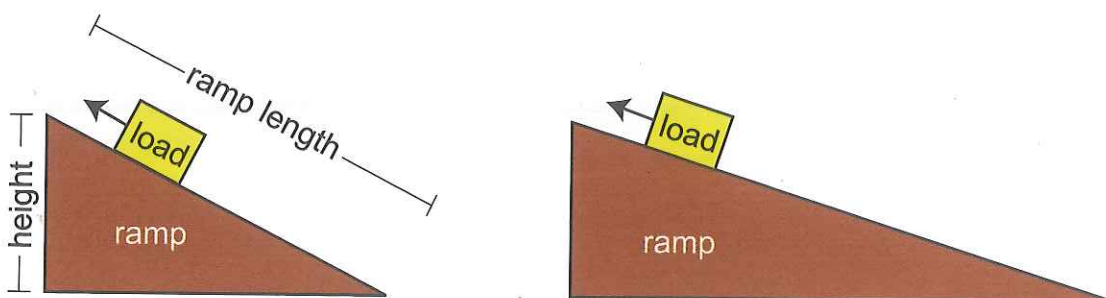
- (A) milk and mud
(B) mud and tea
(C) salt and tea
(D) salt and sugar

21. Which of these did Cali have to keep the same to be able to compare her results?

- (A) the spoon she used to stir the mixture
(B) the size of the glass to hold the mixture
(C) the amount of substance added to the water
(D) the colour of the substance added to the water

22. Tim made two ramps from the same type of wood. He used them to pull a load up a slope.

The diagrams show his ramps.

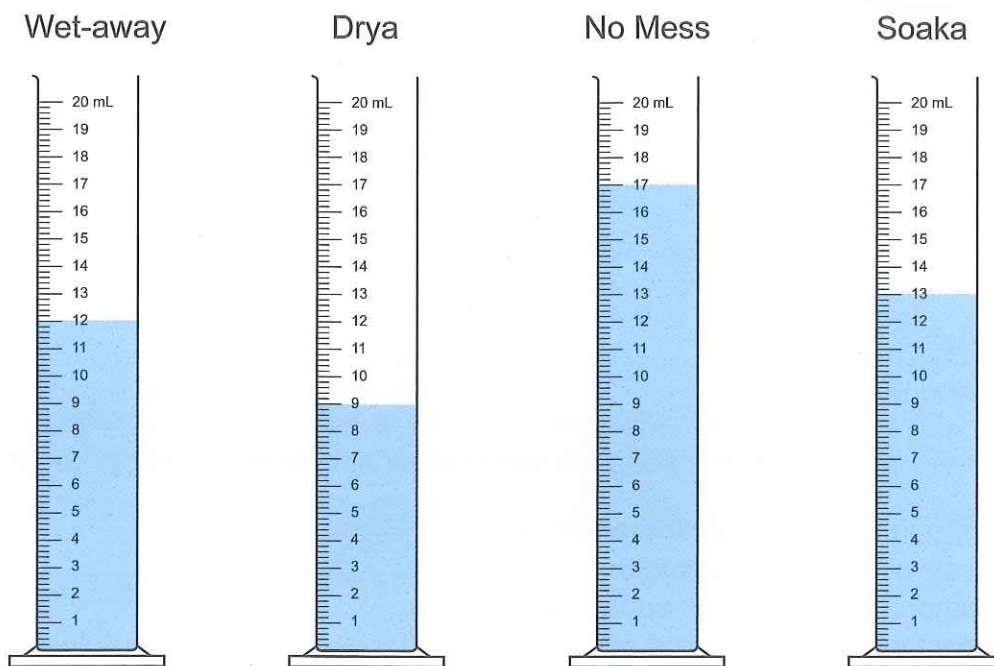


What did Tim change in his investigation?

- (A) the smoothness of the ramp surface
(B) the height the load was raised
(C) the steepness of the ramp
(D) the size of the load

23. Sharon and Andrew wanted to compare four different brands of paper towel. They had equal-sized paper towels of four different brands: Wet-away, Drya, No Mess, Soaka. They had four measuring cylinders, one for each paper towel. Each measuring cylinder contained 20 mL of water.

Sharon and Andrew slowly poured water onto each towel until the towel could not soak up any more. The diagrams show the water left in the measuring cylinders after they had stopped pouring water onto the paper towels.



Which paper towel soaked up the **LEAST** amount of water and how much did it soak up?

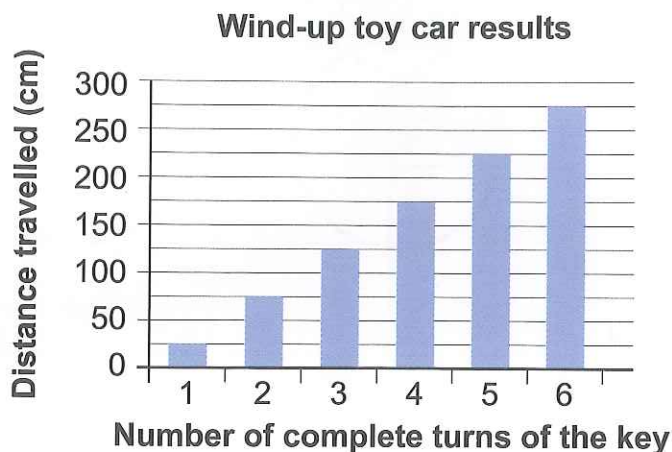
	Paper Towel	Amount of water soaked up (mL)
(A)	Drya	9
(B)	Drya	11
(C)	No Mess	3
(D)	No Mess	17

For questions 24 and 25 use the information below.

Wind-up toy cars move forwards when you wind them up with a key and let them go.

Frank carried out an investigation to find out if the number of times he turns the key affects the distance the car travels.

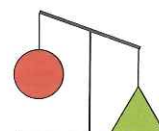
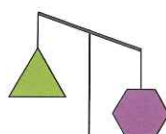
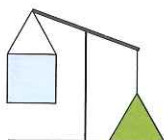
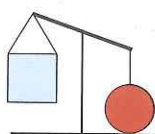
The graph shows Frank's results.



24. What is the least number of complete turns of the key needed to make the car travel 175 cm?
- (A) 3 (B) 4 (C) 5 (D) 6
25. Which two things did Frank have to keep the same to make his investigation fair?
- (A) the car and the distance it travelled
 (B) the car and the surface on which it moved
 (C) the distance the car travelled and the surface on which it moved
 (D) the number of turns of the key and the surface on which the car moved

26. Tina has four different objects. She put them, two at a time, on a balance to see which object was heavier.

The diagrams show the results of her investigation.

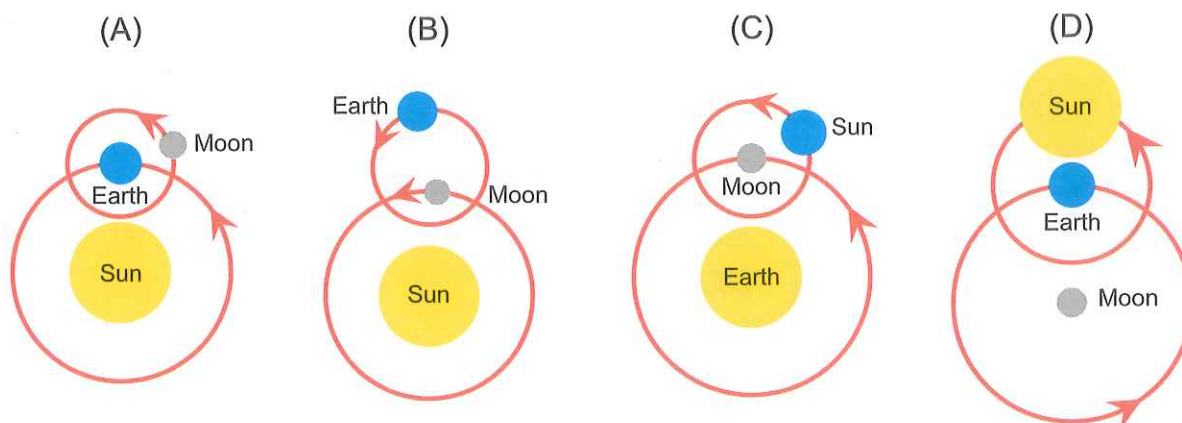


Which option ranks the objects in order from lightest to heaviest?

- (A) (B) (C) (D)
-

27. Earth moves around the Sun. As Earth moves around the Sun, the Moon moves around Earth.

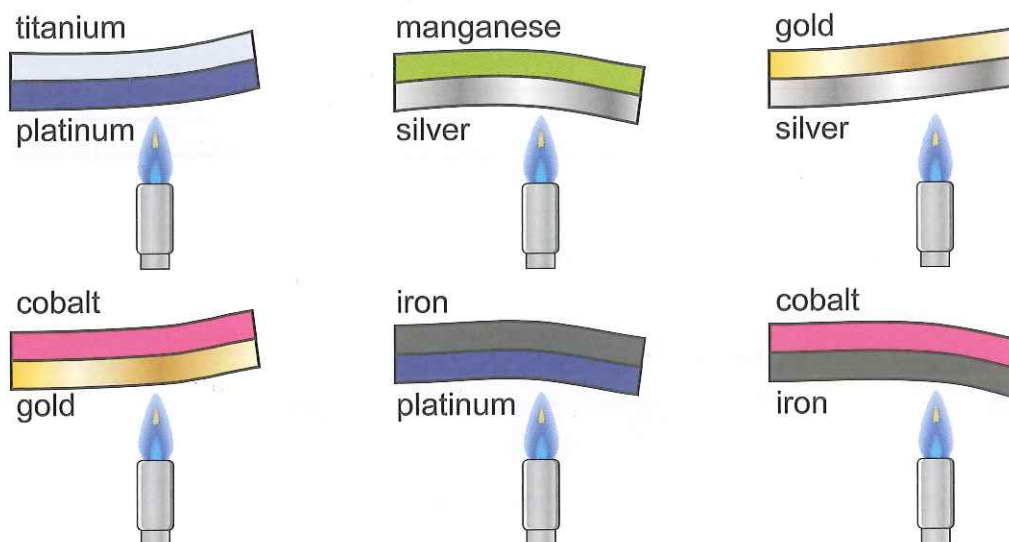
Which diagram best represents the movement of Earth and the Moon?



28. Bimetallic strips consist of two thin strips of different metals bonded together. When the bimetallic strip is heated, one metal expands more than the other and the strip bends.



The diagrams show some bimetallic strips that are being heated.

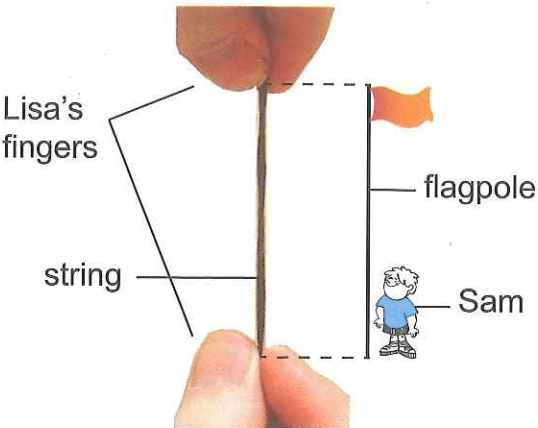
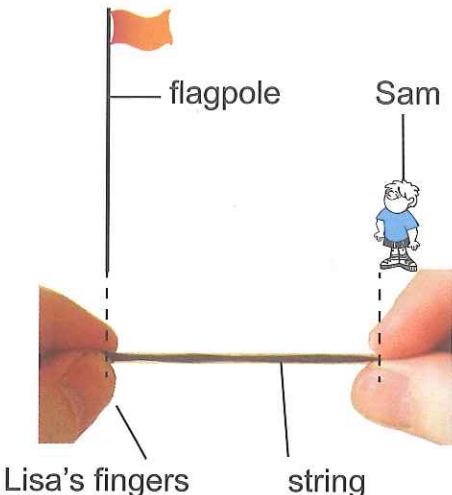


Based on this information, which of these metals expands most when heated?

- (A) titanium (B) manganese (C) cobalt (D) gold

For questions 29 and 30 use the information below.

Sam wanted to find out the height of the flagpole in the school ground. His sister, Lisa, showed him three steps that he could follow to find the answer.

Step 1	Step 2
Lisa told Sam to stand next to the flagpole. Then Lisa stood away from the flagpole and held a piece of string so she could see what is shown below. 	Lisa turned the string. She told Sam to walk away from the flagpole until she could see what is shown below. 

29. What is Step 3?

- (A) measure how tall Sam is
- (B) measure how long the string is
- (C) measure the distance from Lisa to the flagpole
- (D) measure the distance from Sam to the flagpole

30. Which of the following would have the **LEAST** effect on how accurate the measurement is?

- (A) telling Sam to walk behind the flagpole instead of walking to the side
- (B) not holding the string straight when telling Sam where to stop
- (C) moving closer to the flagpole before telling Sam where to stop
- (D) telling Sam to walk to the left of the flagpole instead of walking to the right

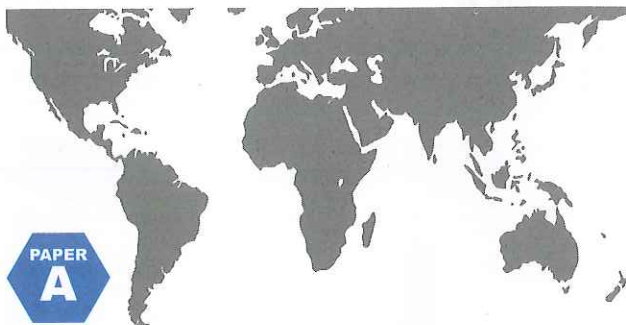
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The following year levels should sit THIS Paper:

Australia	Year 3
Brunei	Primary 3
Indonesia	Year 4
Malaysia	Standard 3
New Zealand	Year 4
Pacific	Year 4
Singapore	Primary 2
South Africa	Grade 3



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- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

FIRST NAME

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☐	☐	☐	☐	☐	☐
A	A	A	A	A	A
B	B	☒	☒	B	E
C	C	C	C	C	C
D	D	D	D	D	D
E	☒	E	☒	☒	☒

LAST NAME

B	A	C	H
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☒	B	E	E
C	C	☒	C
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E	☒	☒	☒

FIRST NAME **LAST NAME**

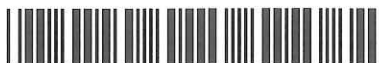
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D	D	○	○
E	○	○	○

A	I	○	B	E	N	G
○	○	○	○	○	○	○
○	A	A	A	A	A	A
○	B	B	○	B	B	B
○	C	C	○	C	C	C
○	D	D	○	D	D	D
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FIRST NAME										LAST NAME				
J	A	M	A	L			B	I	N	A	B	A	S	
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E	E	E	E	E	E	E	E	E	E	☐	☐	☐	☐	

[illegible][illegible]

☐ Male ☐ Female



☐ Yes ☐ No

Town / suburb: _____

Today's date: / / Postcode:

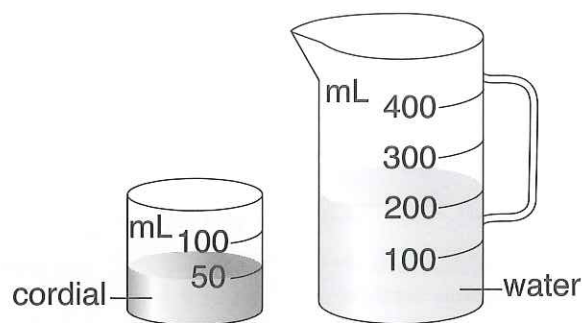
Day		Month		Year	
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1	1	1	1	1	1
2	2		2	2	2
3	3		3	3	3
4			4	4	4
5			5	5	5
6			6	6	6
7			7	7	7
8			8	8	8
9			9	9	9

(A)	(K)
(B)	(L)
(C)	(M)
(D)	(N)
(E)	(O)
(F)	(P)
(G)	(Q)
(H)	(R)
(I)	(S)
(J)	(T)

TO ANSWER THE QUESTIONS

Example: Ari added cordial to water to make a jug of drink.
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval **(D)**, as shown.

(A) (B) (C) ☒



START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
- 3 (A) (B) (C) (D)
- 4 (A) (B) (C) (D)
- 5 (A) (B) (C) (D)
- 6 (A) (B) (C) (D)
- 7 (A) (B) (C) (D)
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- 11 (A) (B) (C) (D)
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- 13 (A) (B) (C) (D)
- 14 (A) (B) (C) (D)
- 15 (A) (B) (C) (D)

- 16 (A) (B) (C) (D)
- 17 (A) (B) (C) (D)
- 18 (A) (B) (C) (D)
- 19 (A) (B) (C) (D)
- 20 (A) (B) (C) (D)
- 21 (A) (B) (C) (D)
- 22 (A) (B) (C) (D)
- 23 (A) (B) (C) (D)
- 24 (A) (B) (C) (D)
- 25 (A) (B) (C) (D)
- 26 (A) (B) (C) (D)
- 27 (A) (B) (C) (D)
- 28 (A) (B) (C) (D)
- 29 (A) (B) (C) (D)
- 30 (A) (B) (C) (D)



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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	D	D	D	B	C	B	D	C	C
2	C	D	C	C	A	D	D	A	D
3	D	B	B	C	D	C	B	B	A
4	B	A	B	B	B	B	B	B	B
5	D	D	D	C	D	C	C	A	B
6	C	D	C	B	D	B	A	C	A
7	A	A	A	D	B	A	C	D	D
8	D	D	C	A	C	D	B	B	D
9	C	A	D	D	B	D	D	C	A
10	C	C	D	B	C	C	A	A	B
11	A	D	A	A	B	B	C	D	A
12	B	A	B	B	D	D	C	A	A
13	C	D	A	C	A	D	A	D	A
14	B	B	D	A	C	C	B	B	B
15	D	C	D	B	D	A	D	B	D
16	C	C	D	C	A	C	D	A	C
17	B	D	A	B	C	D	D	C	C
18	A	D	C	B	B	C	D	C	D
19	A	B	A	C	D	B	B	C	A
20	D	B	C	D	D	D	D	C	C
21	C	C	C	D	C	B	C	A	D
22	C	C	A	D	B	D	A	D	D
23	C	C	A	D	C	D	C	D	D
24	B	D	A	D	B	B	C	C	D
25	B	B	B	C	D	B	D	A	C
26	A	A	A	C	C	C	D	C	D
27	A	A	D	A	A	C	B	B	C
28	B	D	C	A	D	D	C	D	A
29	D	C	A	C	C	A	C	A	D
30	D	A	C	D	C	C	A	C	A

(Please turn over)



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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			C	B	B	D	C	B	C
32			A	C	C	C	B	C	B
33			C	D	D	C	D	C	C
34			B	D	D	C	B	D	D
35			D	D	D	C	B	C	C
36			C	C	A	A	B	A	D
37			D	B	C	D	B	D	B
38			B	D	C	C	A	B	C
39			B	A	A	C	A	C	D
40			B	A	D	D	C	B	D
41					A	D	D	B	A
42					A	D	C	B	A
43					D	A	B	A	D
44					C	C	C	D	B
45					B	B	D	D	A

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